



Linked Infrastructure for
Networked Cultural Scholarship

GIS to LOD and KGs

Mobilizing Decades of HGIS Data



Linked Infrastructure for
Networked Cultural Scholarship

Seminar 2: Data Modelling and CIDOC CRM

Introduce the CIDOC CRM ontology. Work in teams to consider data modelling, guided by research questions.

Agenda

1. Intro & Feedback/Comments
2. Recap
3. Context Explorer Demo
4. Intro to Modelling with CIDOC-CRM
5. Breakout Rooms
6. Wrap Up



In-Person Workshop @ University of Ottawa

Date/Time	Title	Description
Afternoon of April 15	Towards a Canadian History Shared Model	Modeling geographic data; focus on interoperability
Full day of April 16	Implementing the Modeling with the Data	Using Claude Code to transform the 4 datasets
Morning of April 17	Connecting HGIS LOD to Canadiana.ca	Stakeholders Panel Exploring future possibilities Reflection on learnings, implications, challenges Future steps



Goals of this Project

- Create LOD, lifting up reliable but underrepresented data and reducing data silos
- Find ways to responsibly incorporate LLMs in the process of creating high-quality LOD



RECAP: URIs and PIDs

URL: the address of a web page

<https://www.saskatoon.ca/>

URI: a unique identifier for something, often but not always a web address

PID: An identifier that there is a reasonable hope will continue to be maintained

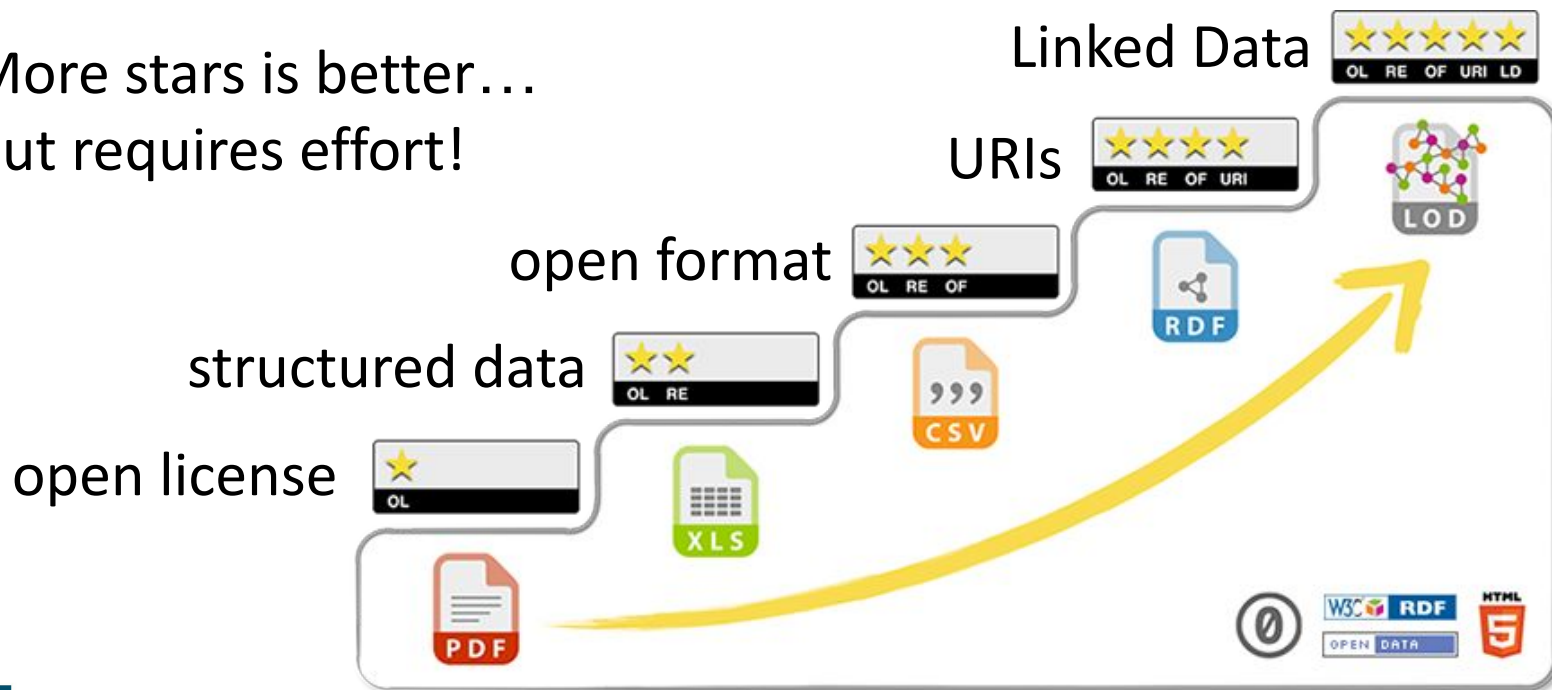
<https://www.geonames.org/6141256/saskatoon.html>

For LOD purposes, URIs and PIDs should resolve on the web to information (human- *and* machine-readable) about what they identify.



RECAP: 5 Stars of Linked Open Data

More stars is better...
but requires effort!



RECAP: Interoperability



LOD: silos with transparency & bridges



Feedback

- How was this?
- Was anything confusing?



Context Explorer





Government
of Canada

Gouvernement
du Canada

Français

Collection search

Sign in

[Canada.ca](#) > [Library and Archives](#) > [Collection](#) > [Search the collections](#)

Two letters from Frederick Douglass to Mary Ann Shadd Cary concerning a tour of the Reconstructionist South.

CONTENTS

Thumbnails

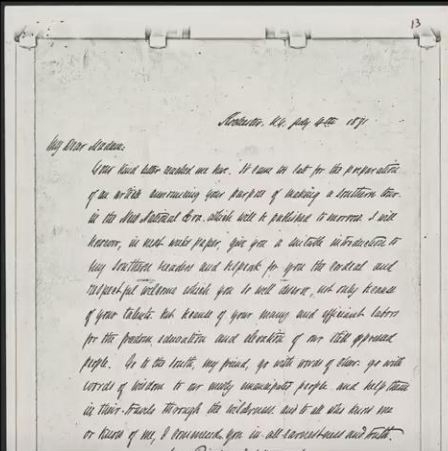
Details





e011536884-007

Two letters from Frederick Douglass to Mary Ann Shadd Cary concerning a tour of the Recons...



Intro to Modelling with CIDOC-CRM



Places

and their complexities

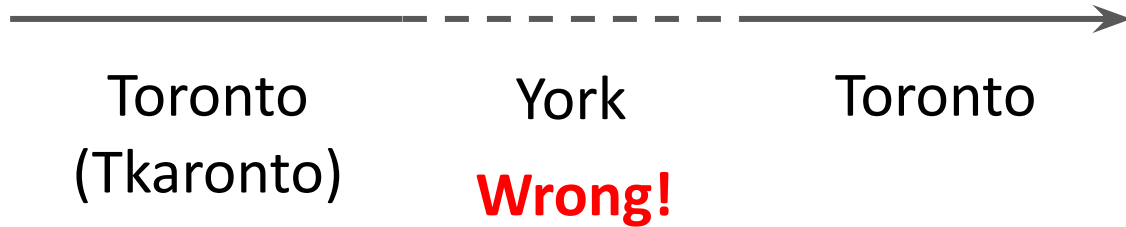


Toronto



Toronto

- Toronto is an Indigenous name (from Kanien'kehá:ka term *Tkaronto* = “tree in the water”)
- Fort Toronto/Fort Portneuf constructed by the French in 1750
- From 1793 to 1834 the British referred to the city under the name “York”



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On the other side of the ocean...

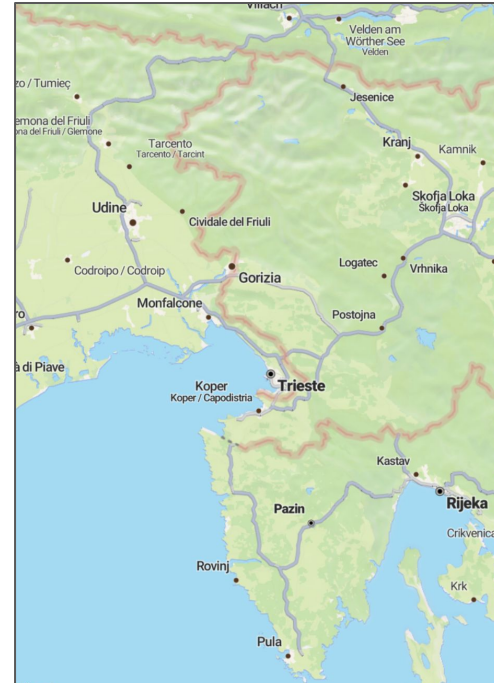


On the other side of the ocean...



A contested land across centuries

- ↑
- 1991 → Slovenia / Croatia
 - 1947 → Yugoslavia
 - 1919 → Kingdom of Italy
 - 1814 → Austrian Empire
 - 1805 → France
 - 1797 → Holy Roman Empire
 - 1348 → Republic of Venice



Moving cities

Oslo, Norway

- Founded in 1048
- Destroyed by a fire in 1624
- Rebuilt in a different location under the name Christiania
- Then it grew larger
- Renamed Oslo again in 1925



Lost places

The land of Punt

- Somewhere in East Africa
- Well known to Ancient Egyptians
- No one knows exactly where it was



What is a place?

Many different things, for example...



City

location, administrative entity, group



Theatre

location, building, organization



Lake

location, physical feature, ecosystem



Many complexities

- Names changing over time
- Locations changing over time
- Areas changing over time
- Borders moving
- Contested histories
- Historical uncertainties
- Nature of place itself



Place Data



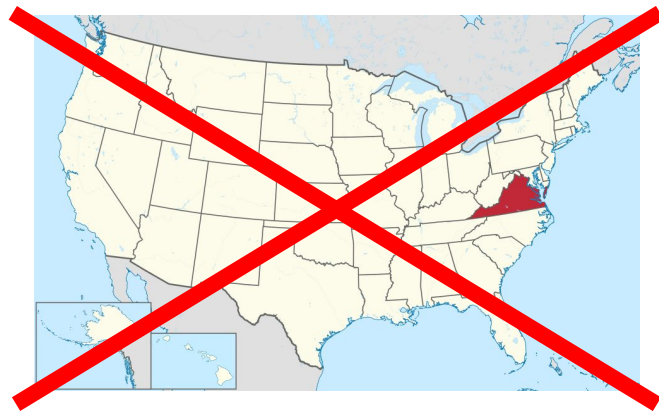
More tables

Entity
Virginia



More tables

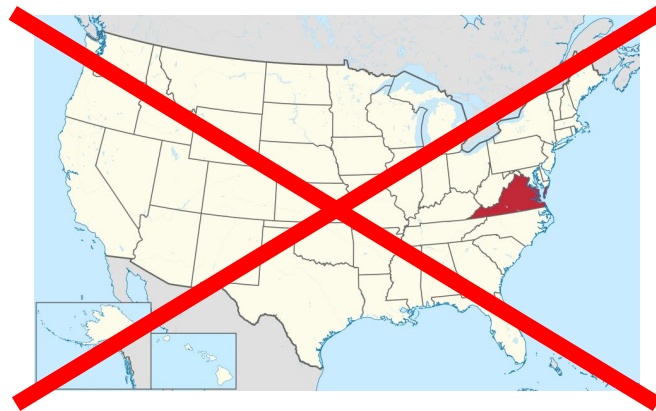
Entity	Place
Virginia	London



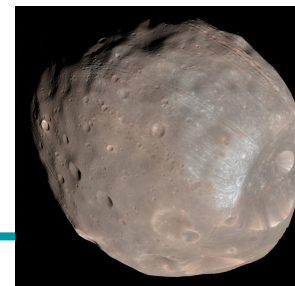
More tables

Entity	Place
Virginia	London

Person?



Place?



More tables

Entity	Place
Virginia	London

Virginia
Woolf



London,
England



Ambiguities

Entity	Place
Virginia	London

Some ambiguity is found in the data...

...and you could resolve it using URIs



Ambiguities



Entity	Place
Virginia	London

But what do these headings mean?

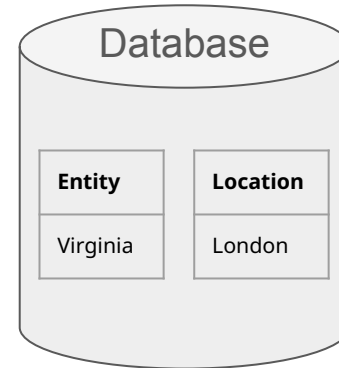
And what is the relation between them?



Not a problem of structure...

Entity	Place
Virginia	London

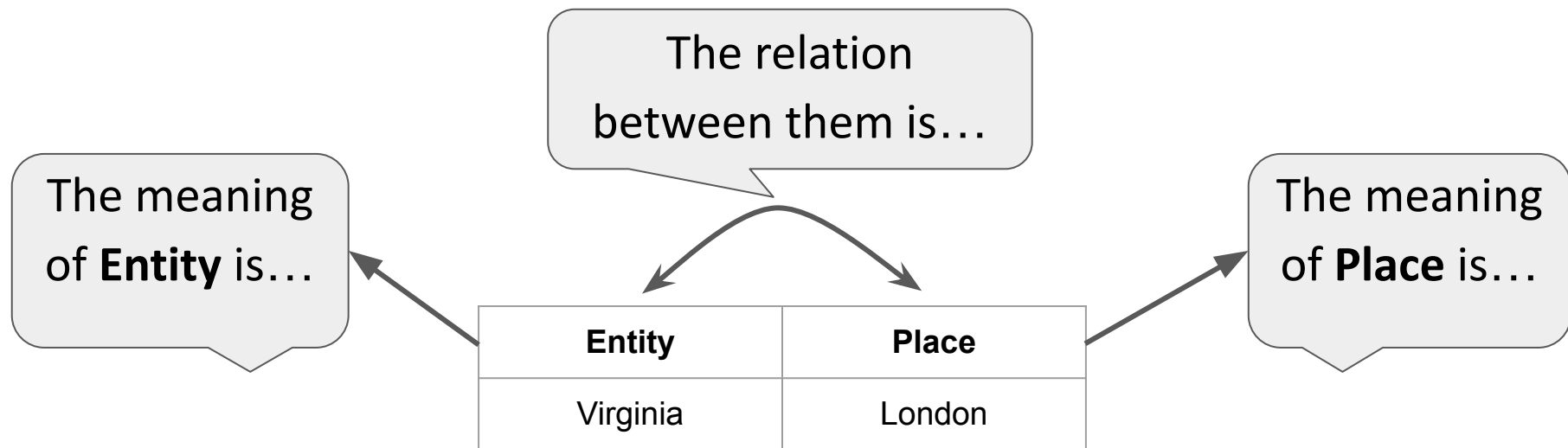
```
{  
  "Entities": [  
    "Virginia":  
      "Location": "London"  
  ]  
}
```



A problem of meaning



Wouldn't it be nice if...



Ontology Modelling



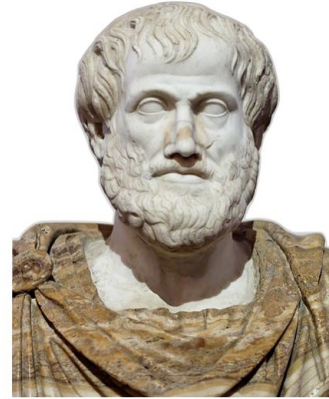
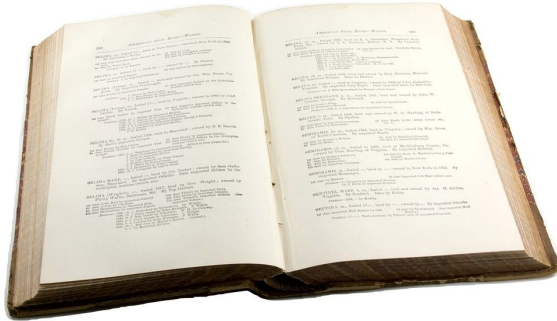
What is an ontology?

Etymology of the word

From Greek

ontos, “being”

logia, “study of”



In philosophy,
“the study of being”
(discussed by Aristotle
in the *Metaphysics*)



OK, but what is an ontology?

In computer science, “**ontology**” means...

“A formal, explicit specification of a shared conceptualisation”



Guarino, Oberle & Staab (2009) “What is an ontology?”

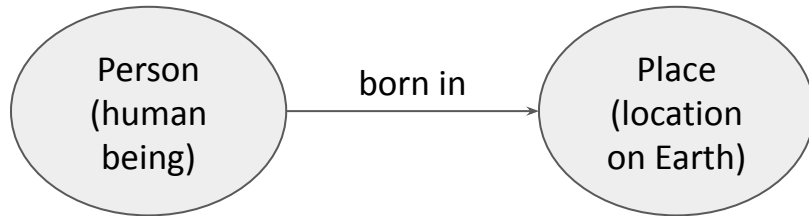
Studer, Benjamins & Fensel (1998) “Knowledge engineering: Principles and methods”



Conceptualisation

“A **person** is a human being. A **place** is a location on the surface of the Earth. A person is **born in** a place.”

“A formal, explicit specification of a shared **conceptualisation**”



Shared Conceptualisation

“A **person** is a human being. A **place** is a location on the surface of the Earth. A person is **born in** a place.”

“A formal, explicit specification of a **shared conceptualisation**”



From conceptualisation to ontology

“A **person** is a human being. A **place** is a location on the surface of the Earth. A person is **born in** a place.”



“A **formal, explicit specification** of a shared conceptualisation”

```
:Person a owl:Class ;  
    rdfs:comment "a human  
        being" .  
:Place a owl:Class ;  
    rdfs:comment "a location  
        on Earth" .  
:born_in a rdf:Property .  
    rdfs:domain :Person ;  
    rdfs:range :Place ;  
    rdfs:comment "connects  
        a person to the place  
        they were born in"
```



OWL (Web Ontology Language)

- [OWL](#) is the **ontology language** of the **Semantic Web**
- **Open standard** approved by [W3C](#) (World Wide Web Consortium)
- **Formally** encodes meaning, based on description logics
- Expressed using **RDF (Resource Description Framework)**



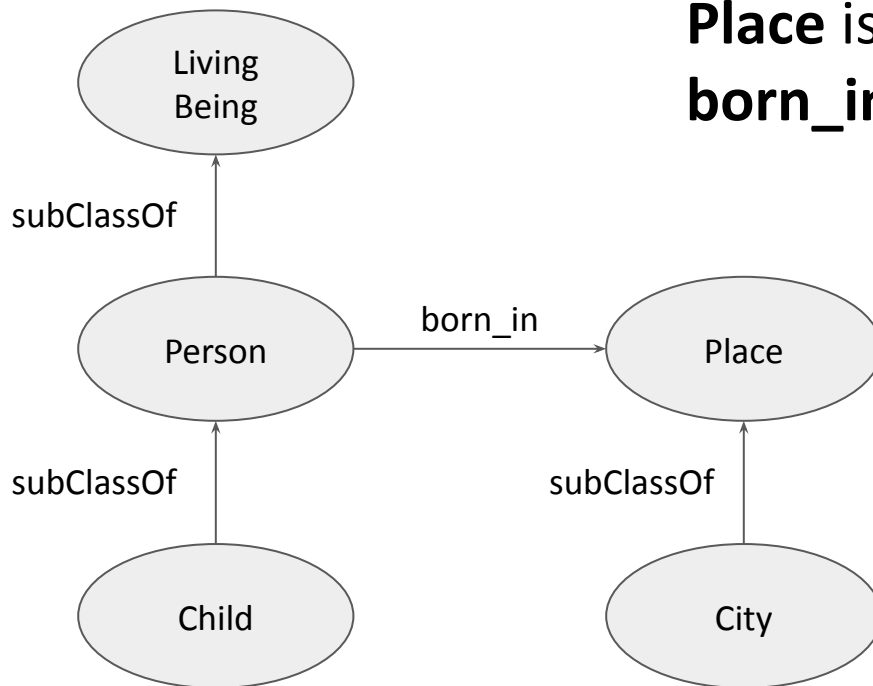
Classes and properties

Person is a class

Place is a class

born_in is a property

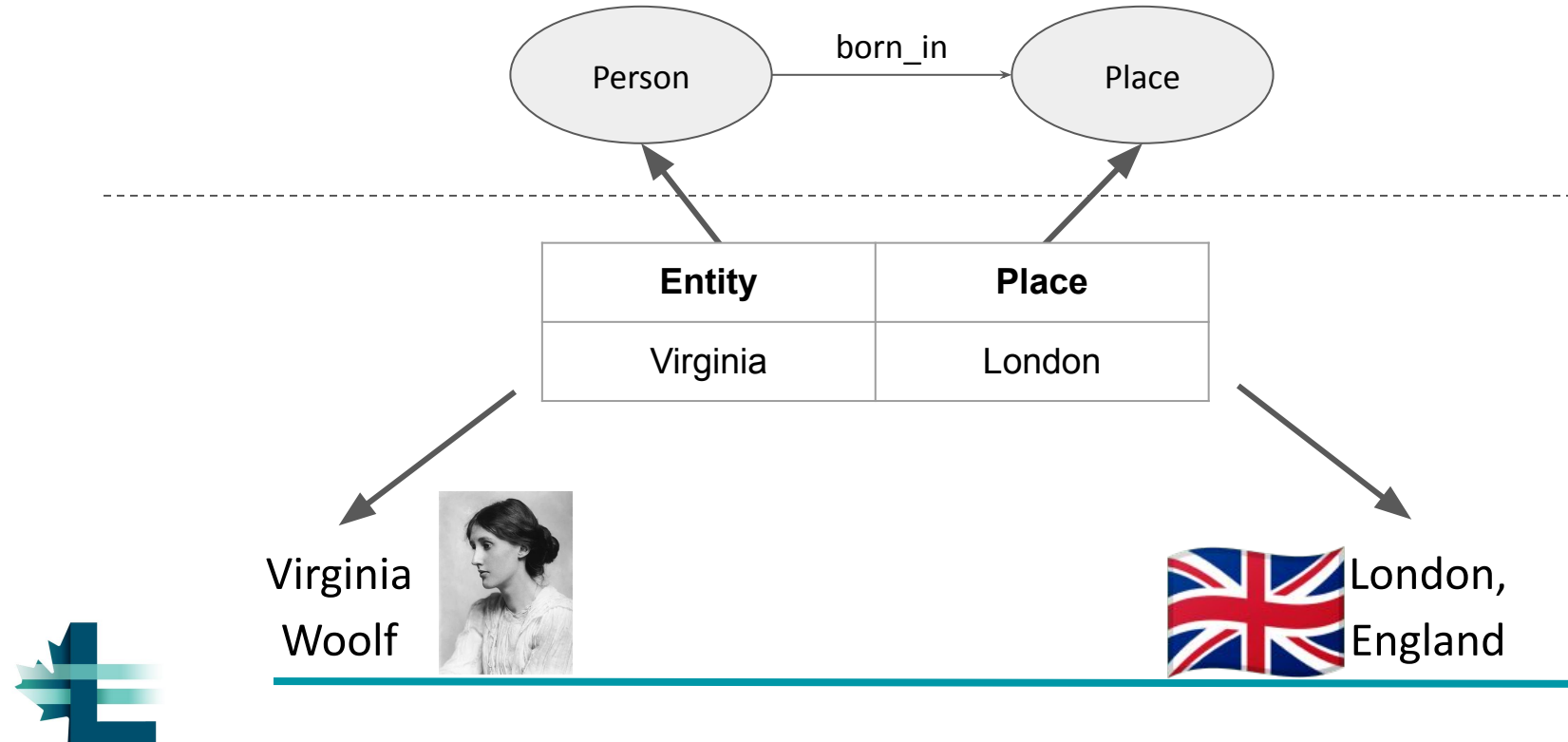
*This is
the
model*



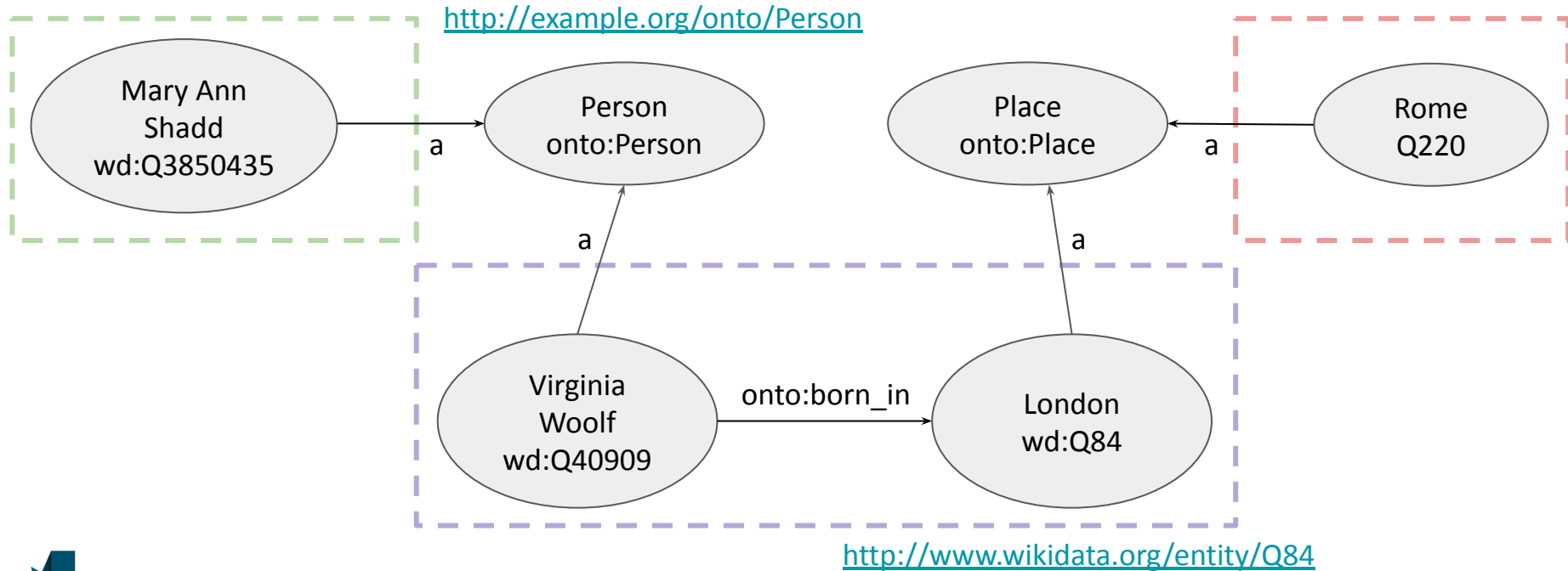
*Where
is the
data?*



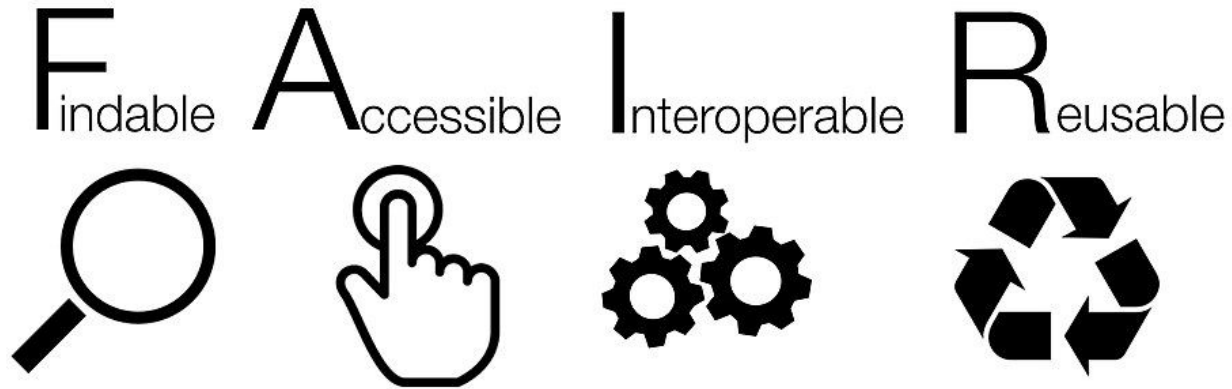
Ontology mapping



Knowledge graph



Ontologies make data FAIR

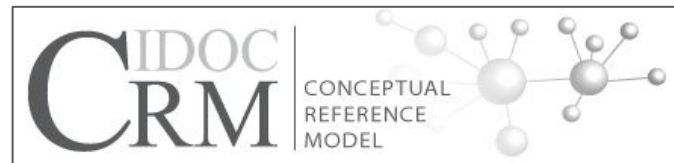


CIDOC CRM

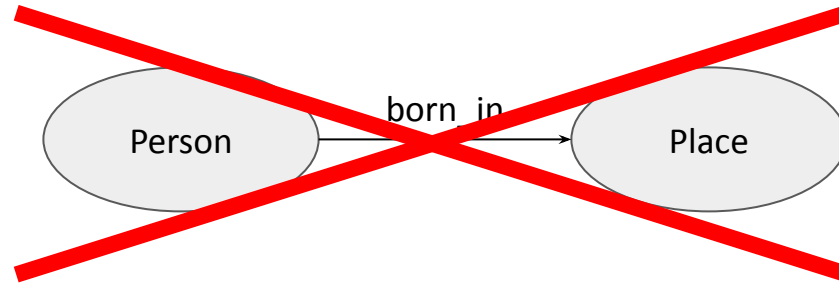


Our reference ontology

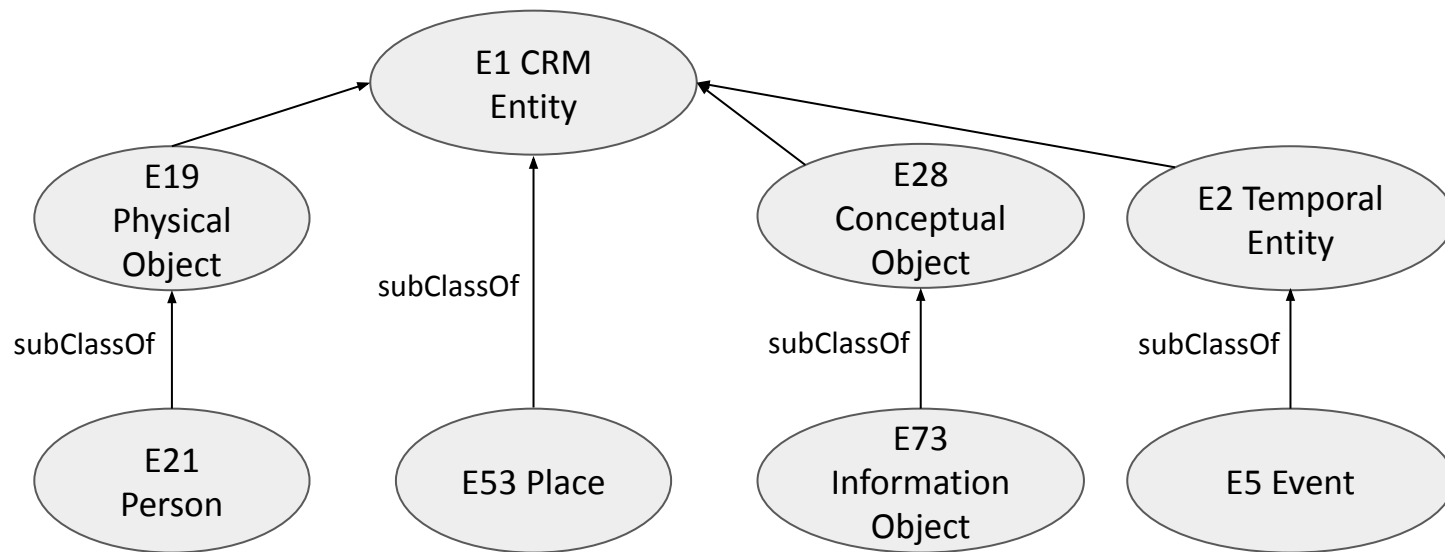
- **CIDOC CRM** is a standard ontology
- Created by **CIDOC** (International Committee for Documentation) of **ICOM** (International Council of Museums)
- **CRM** = Conceptual Reference Model
- General enough to applied across different domains



CIDOC CRM is event-based



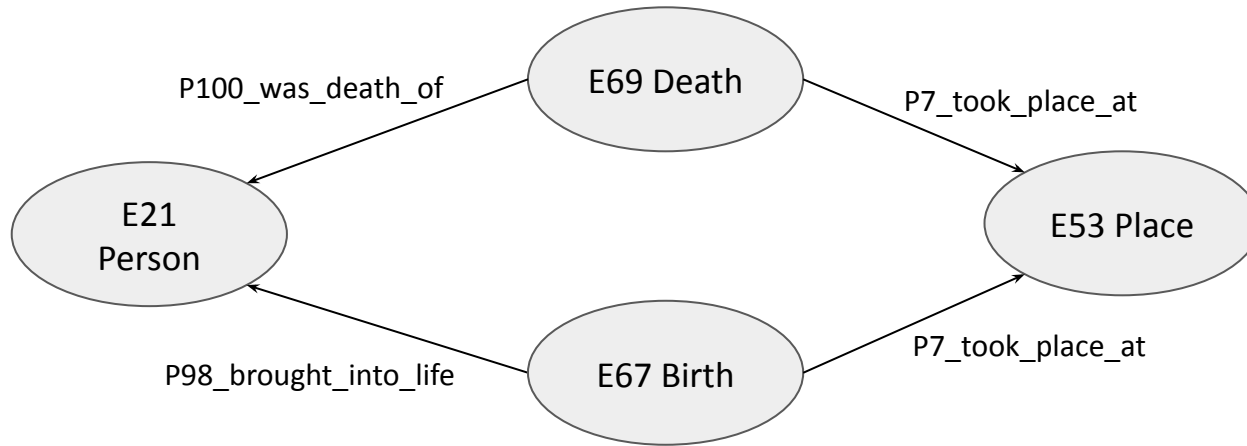
CIDOC CRM classes



...and many more (almost 100)



CIDOC CRM properties

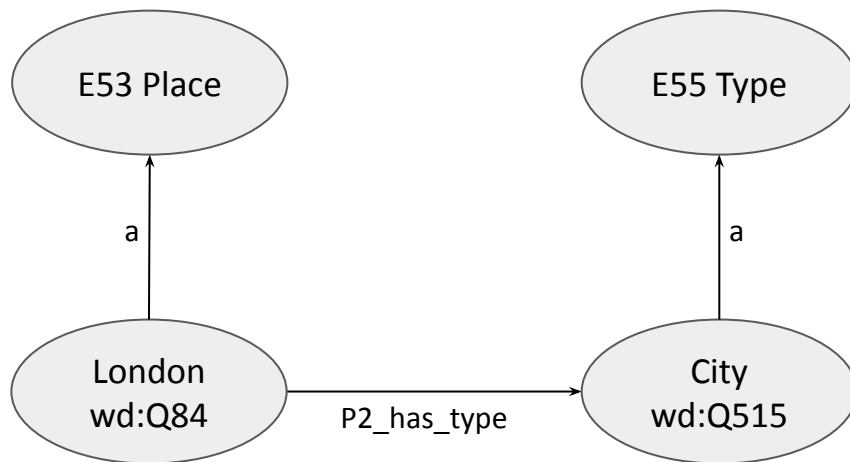


...and many more (almost 200)



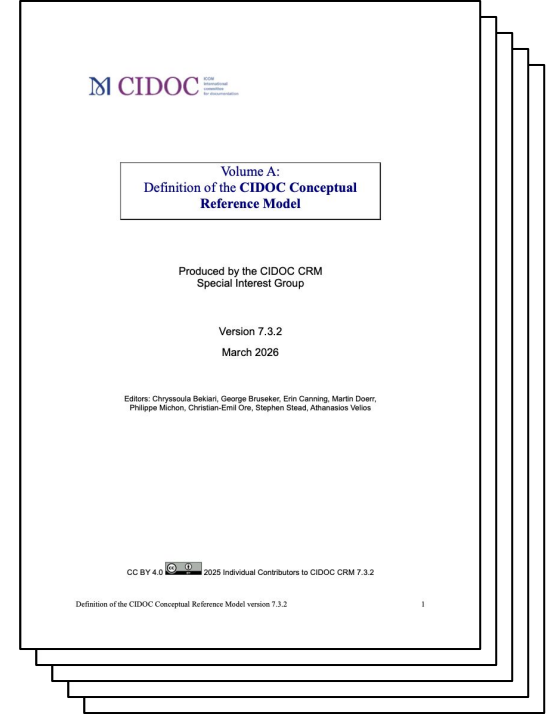
Using External Vocabularies

You want to model a city but there is no specific class for cities. Rather than creating new classes, you use a special property P2.

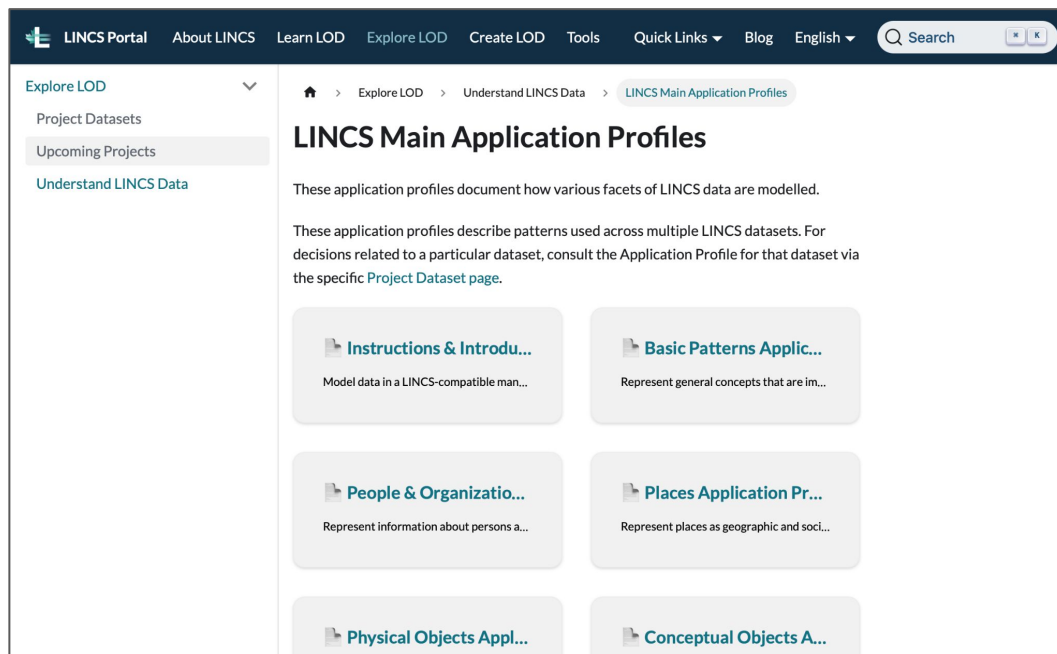


Documentation

- CIDOC provides extensive documentation
- https://cidoc-crm.org/sites/default/files/cidoc_crm_version_7.3.2.pdf
- 243 pages long 🤖
- 🤖🤖🤖



LINCS Application Profiles



The screenshot displays the LINCS Portal interface. The top navigation bar includes links for 'LINCS Portal', 'About LINCS', 'Learn LOD', 'Explore LOD', 'Create LOD', 'Tools', 'Quick Links', 'Blog', and 'English'. A search bar is located on the right. The left sidebar shows a dropdown menu for 'Explore LOD' with options: 'Project Datasets', 'Upcoming Projects', and 'Understand LINC Data'. The main content area is titled 'LINCS Main Application Profiles' and contains the following text:

These application profiles document how various facets of LINCS data are modelled.

These application profiles describe patterns used across multiple LINCS datasets. For decisions related to a particular dataset, consult the Application Profile for that dataset via the specific [Project Dataset page](#).

Below the text are six application profile cards arranged in a 3x2 grid:

- Instructions & Introdu...**
Model data in a LINC-compatible man...
- Basic Patterns Applic...**
Represent general concepts that are im...
- People & Organizatio...**
Represent information about persons a...
- Places Application Pr...**
Represent places as geographic and soci...
- Physical Objects Appl...**
- Conceptual Objects A...**



Places in CIDOC CRM



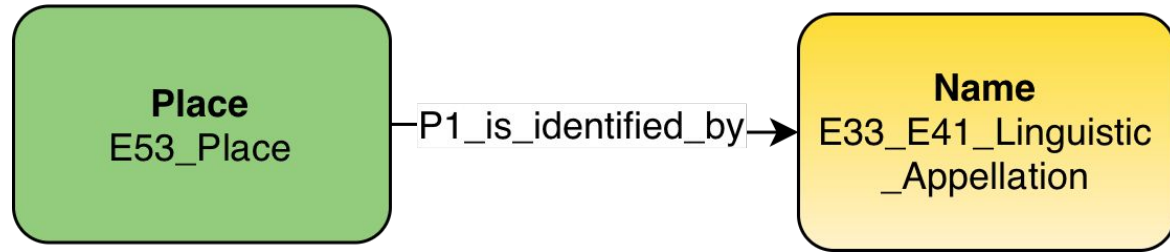
The Place class

“This class comprises extents in the natural space we live in, in particular on the surface of the Earth, in the pure sense of physics: independent from temporal phenomena and matter. They may serve to describe the physical location of things or phenomena or other areas of interest. Geometrically, instances of E53 Place constitute single contiguous areas or a finite aggregation of disjoint areas in space which are each individually contiguous. They may have fuzzy boundaries.”

Place
E53_Place



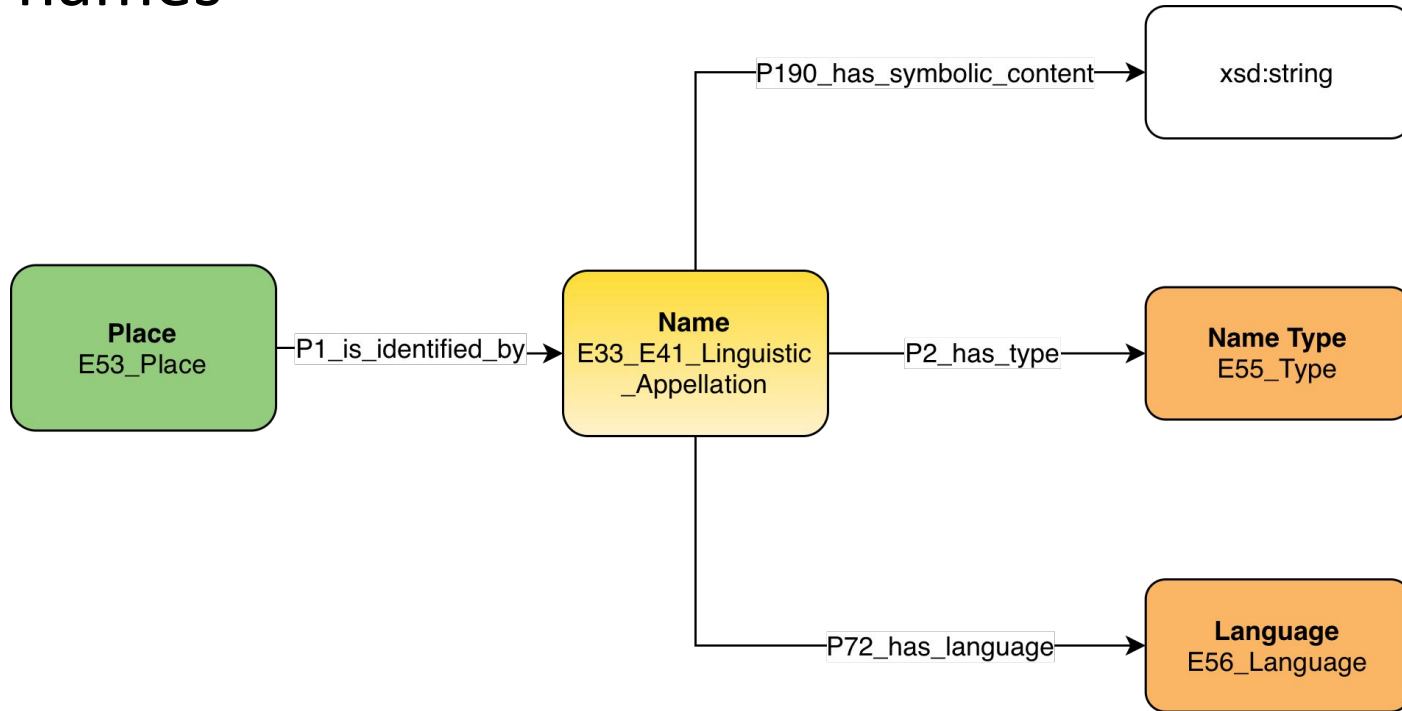
Place names



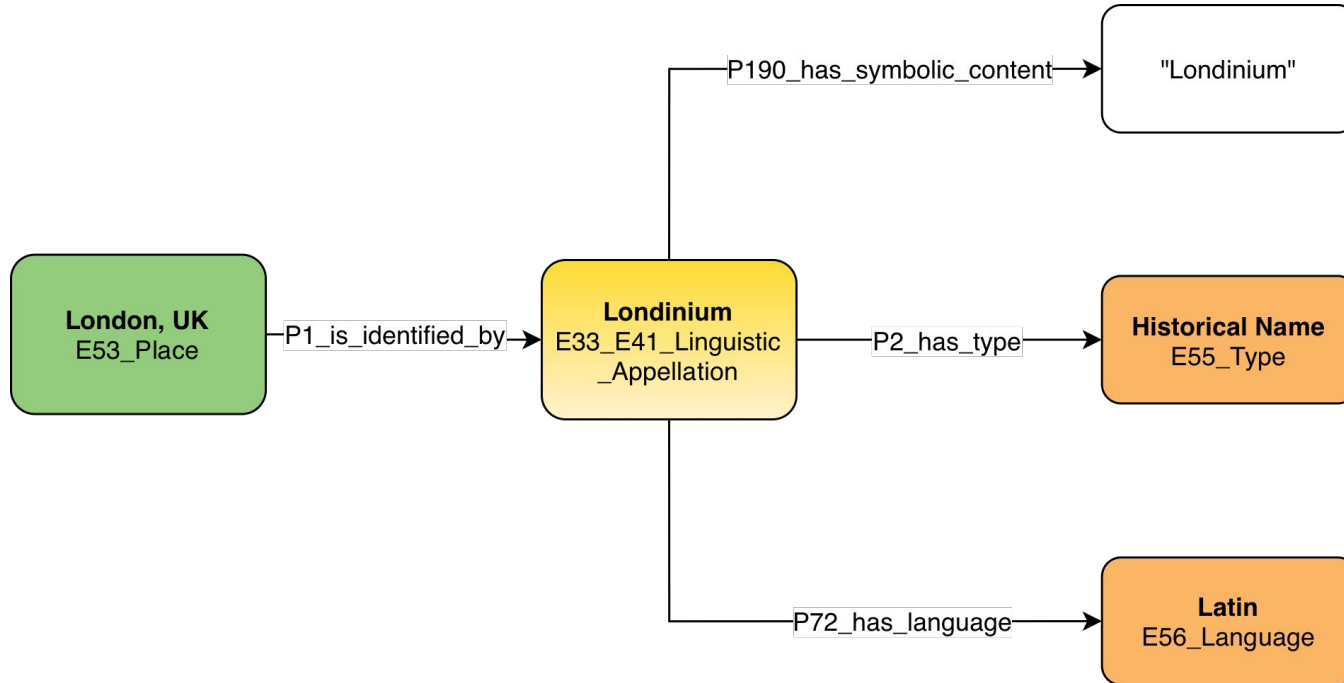
Place names are modelled as **linguistic appellations**



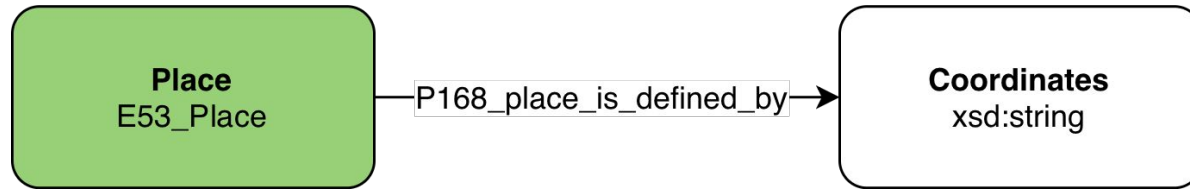
Place names



Place names



Place coordinates

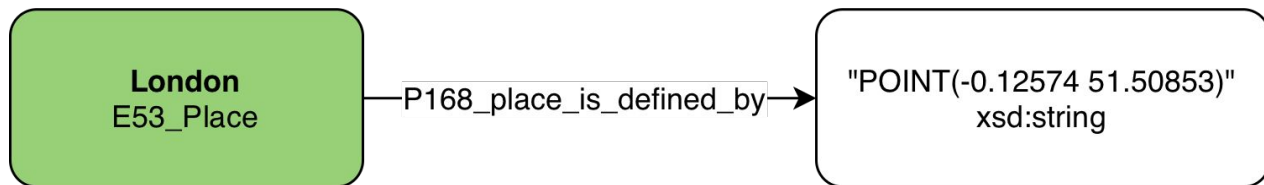


Place coordinates are modelled as **space primitives**

In practice, these are often strings of text



Place coordinates

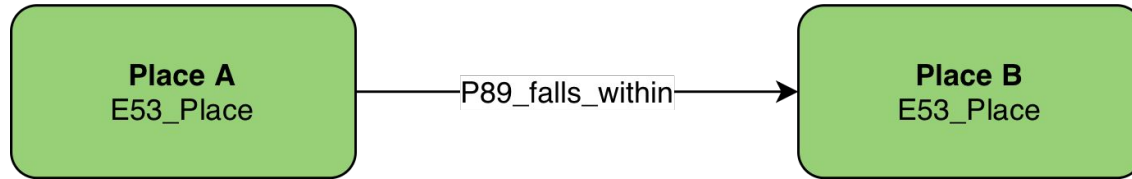


Place coordinates are modelled as **space primitives**

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Relations among places

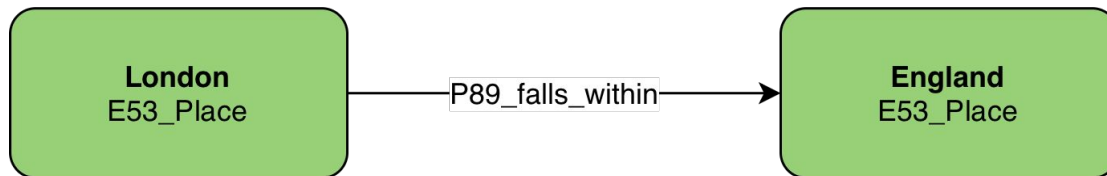


P89 falls within is commonly used, but also

- *P121 overlaps with*
- *P122 borders with*
- *P189 approximates*



Relations among places



P89 falls within is commonly used, but also

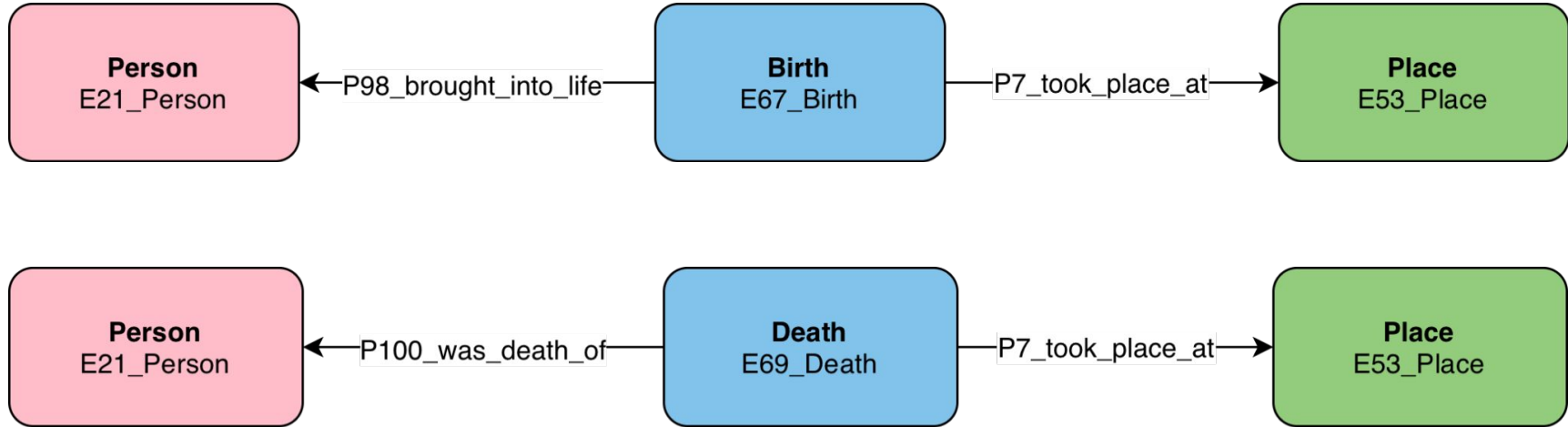
- *P121 overlaps with*
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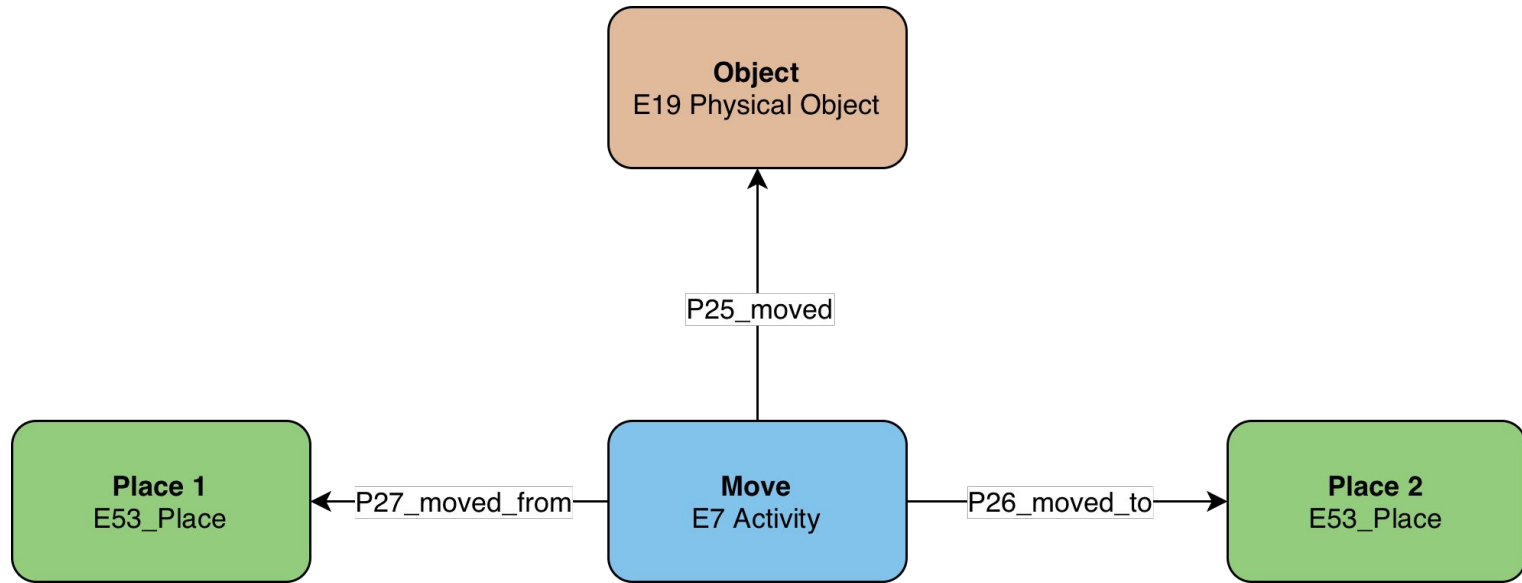
Places and events



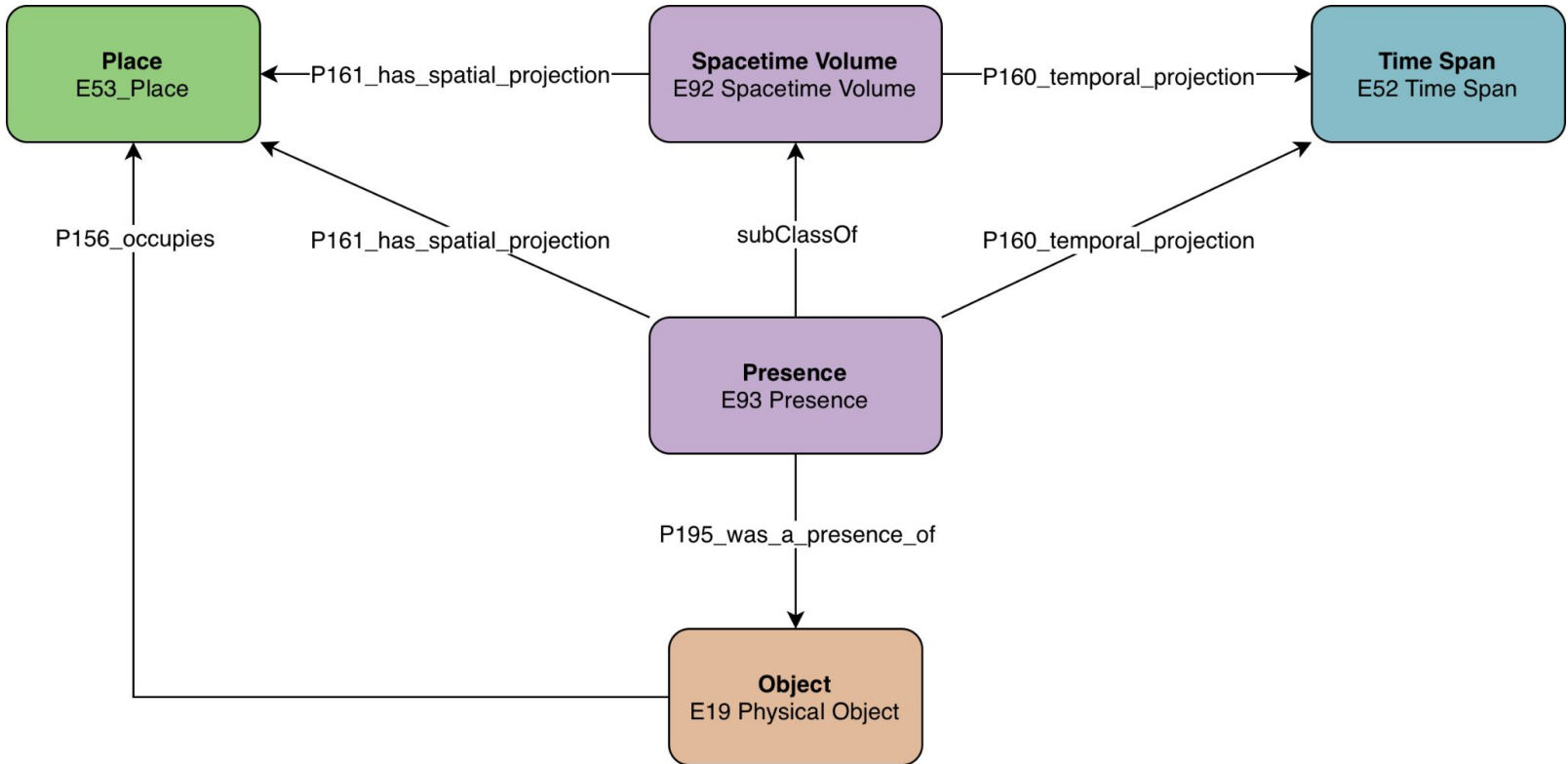
Places and people



Places and physical objects



Places over time



For more information...

- [LINCS main application profiles](#)
 - Especially [Places application profile](#)
- LINCS Canadian History application profiles
 - [Historical Canadian Persons](#)
 - [Historical Indian Affairs Agents](#)
 - [Cabinet Conclusions](#)
- [CIDOC CRM documentation](#)



5 Minute Break



Use Cases

Developing Research Questions for Linked Data

Defining the Data's Potential

- Assess what the data currently answers vs. what it could answer once connected.
- Identify "data gaps": What connections or external sources are missing to reach your goal?
- Determine how to quantify or measure information that is currently qualitative.

Framing the Inquiry

- The Vision: What do you wish your data could reveal?
- The Perspective: How do questions change when viewed by a researcher, a student, or the general public?
- The Format: What does a "successful" answer look like (e.g., a specific visualization, a list, or a map)?



Why

- Use these questions to inform the requirements of the data model
- Bridge the gap between natural language and code where possible
 - what can reasonably be standardized?
 - what needs to remain qualitative to provide rich context?
- Bridge across domains
 - broaden perspective on your data to allow for comparison and interoperability
- Ensure the data model supports the specific visualizations needed to tell your data's story



Breakout Rooms

Please take notes!



*Your group today doesn't have
to be final choice for Ottawa*

Dataset Activity

Room 1:

Canadian Census data -
Census Subdivisions

Room 1:

Canadian Census data -
Farm microdata

Room 2:

Montreal lots and owners 1725

Room 4:

Métis communities

[Links to sample data](#)



Example use cases

- Discoverability

- What are all of the primary sources in *Canadiana* that mention the Métis communities or settler communities within 10km of a Métis community?
- Which Montréal land owners appear in published sources from the early 18th century held by Canadiana or BANQ?

- Analysis

- Which Montréal landowners have *DCB* entries. Who are they connected to in their biographies; are those people in the landowner database?

- Future - GraphRAG

- LLM assisted search using a Knowledge Graph + Unstructured Text Vector Database
- What do primary sources in Canadiana reveal about economic life in Métis communities after 1885?



Review the data sample:

- 1) What data cleaning work is needed?
- 2) What work is needed to add URIs to the data?
- 3) What would be interesting to do with the data as LOD? What research questions could LOD potentially answer
- 4) What would be challenging about modelling it as LOD?

What would be challenging?

Could LLMs help?

Could LLMs fail and how?



Wrap Up



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Homework

Think about:

- *What aspects of your dataset could we reconcile to Geonames or Wikidata?*
 - *What is the scale of this work? Can LLMs help?*

Introductory Linked Open Data Readings:

- Posner, [Introduction to Linked Open Data](#) (Youtube)
- Blaney, [“Introduction to the Principles of Linked Open Data”](#)
- Crompton, Linked Open Data: Understand It, Use It, Make It! [Part 1](#) & [Part 2](#) (Youtube)
- Programming Historian, [Introduction to the Principles of Linked Open Data](#)
- Berners-Lee, [The Semantic Web](#), Scientific American

Optional: Claude Code setup Session on April 10th



Optional: Claude Code Setup Session - April 10th

- Basic Introduction to Claude Code
 - LLM in your command line/terminal that writes code based on English/French instructions.
- Introduction to the Wikidata Model Context Protocol (MCP) server for reconciliation
- Creating Visualizations to confirm data and bespoke Human in the Loop verification websites
 - <https://jburnford.github.io/indianofficelist/review/>
 - <https://jburnford.github.io/Canada-History-Knowledge-Graph/>

